# Office of Research Assurance

Human Subject Research Protections Standard Operating Procedures (SOP)

## SOP - 950: Scientific Merit Review

| Number | Effective Date | Version Number | Previous Versions |
|---|---|---|---|
| 950 | 5/21/2024 | v.1 | n/a |

### Purpose

The purpose of HRP – 950 is to inform researchers, NASA IRB staff, and NASA IRB members when scientific merit review is required prior to NASA IRB review and approval.

Regulatory IRB approval criteria, specifically 14CFR1230.111(a), require that risks be minimized through procedures that are consistent with sound research design and that potential risks of the research are reasonable in relation to anticipated benefits. Scientific merit review substantially contributes to IRB confirmation of investigational rigor, adequacy of the proposed methods to accomplish research objectives, and the feasibility of the planned research when determining risk to benefit ratio of the proposed activity.

## Policy

1. All non-exempt human subjects research requires a scientific merit review (or equivalent) prior to NASA IRB review and approval.
    a. For NASA-funded studies, the Principal Investigator (PI) must upload a scientific merit review letter from the Program Manager or Element Scientist.
    b. For other US government agency funded or national foundation funded studies, the PI must upload a scientific merit review or equivalent letter from the Chief Scientist or relevant party. Often, the peer-review process will suffice.
    c. For studies funded by non-US government agencies, the PI must upload a scientific merit review letter or equivalent from the relevant party.
    d. For studies initiated by International Partner (IP) Agencies (JAXA, CSA, ESA, Russia), the PI must upload a letter from the IP Agency indicating the study has been approved.
    e. For studies initiated by other international organizations, the PI must upload a scientific merit review letter or equivalent from the relevant party.

    f. For studies that do not fit into any of the above categories, the NASA IRB Chair or designee will determine what documentation will serve as a scientific merit review on a case-by-case basis.
2. IRB submissions requiring scientific merit review or equivalent must attach documentation of a completed review in the eIRB as outlined above as part of the IRB application.
3. Scientific merit review or equivalent documentation must include reviewer comments where applicable.
4. If a study has been modified after the completion of a scientific merit review or equivalent, the NASA IRB Office may request another scientific merit review or equivalent is completed prior to IRB review and approval. Modifications that would lead to this request include but are not limited to:
    a. A change in study purpose
    b. A change in study aims and/or hypotheses
    c. An addition of a new study arm or procedure
    d. A change that increases risk to participants
5. The NASA IRB Office reserves the right to request additional documentation if deemed necessary to calculate a risk to benefit analysis.
6. Scientific merit reviewers must not have a conflict of interest (financial or nonfinancial).
7. Scientific reviewers must be independent of the proposed study (i.e., not a member of the study team).
8. For studies in which the NASA IRB cedes review to an external IRB, the external IRB of Record may waive the requirement to submit documentation of a scientific merit review or equivalent as part of their IRB process.

## References

1. 14CFR1230

| Version # | Version Notes: | Originated By: | Approved By: | Date Approved: |
|---|---|---|---|---|
| v.1 | Original | S. Asgar/ J. Kisenwether | M. Covington | 21May2024 |
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